

An On-Orbit Star Tracker Benchmark for Resident Space Object Detection and Attitude Estimation

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An On-Orbit Star Tracker Benchmark for Resident Space Object Detection and Attitude Estimation

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Abstract

Space situational awareness increasingly relies on optical observations to detect and track resident space objects and to estimate spacecraft attitude. Many existing resources are synthetic or restricted, and few provide on orbit, wide field of view imagery with joint labels for space objects and stars. We present a dataset of near-infrared images acquired by the Fast Auroral Imager on the CASSIOPE spacecraft between January and August 2023. The collection comprises 1,378 frames with astrometrically calibrated stars and 4,237 manually verified resident space object instances across 160 transits, accompanied by spacecraft ephemeris, attitude, and image quality metrics. We describe the acquisition conditions, calibration and annotation pipeline, and perform technical validation of pointing stability, astrometric accuracy, annotation reliability, and background characteristics. The dataset supports tasks such as resident space object detection in dense star fields, multi-object tracking under realistic orbital motion, and attitude estimation from star tracker class imagery, and is intended as a shared resource for space situational awareness and navigation studies.

Background & Summary

Space activities have accelerated rapidly over the past decade, driven by the miniaturization and commercialization of spacecraft development¹. With the space sector trending toward the deployment of mega-constellations such as Starlink, OneWeb, Guowang, and Qianfan, Earth's orbital environment has become increasingly competitive, contested, and congested¹. In addition to active satellites, numerous inactive artificial objects remain in orbit; collectively, these are referred to as resident space objects (RSOs). The inactive population consists primarily of debris generated from intentional or unintentional fragmentation events^{2,3}, dormant satellites, and rocket bodies. Even small fragments pose a significant threat to operational spacecraft due to their hyper-velocities⁴. Without timely collision-avoidance maneuvers, conjunction events can result in catastrophic collisions, prematurely terminating missions and generating debris clouds that persist in orbit for decades¹. One prominent example is the 2009 Iridium–Cosmos collision, with debris that remains in orbit and continues to endanger RSOs⁵. These risks highlight the critical importance of tracking, monitoring, and maintaining updated catalogs of RSOs as a central component of Space Situational Awareness (SSA).

Several space agencies, national defense organizations, and commercial companies maintain networks of dedicated sensors for SSA⁶⁻⁹. While highly capable, these systems require substantial resources, and their data are often restricted due to overlaps with military applications. Commercial services typically operate on a subscription basis¹⁰⁻¹², limiting accessibility for the wider research community. This creates a significant gap, leaving many researchers reliant on curating their own datasets.

One promising alternative source of SSA data is the use of onboard star-trackers¹³⁻¹⁶, which are commonly flown on satellites for attitude determination. Star-trackers provide wide field of view (FOV) compared to dedicated SSA sensors, enabling them to capture larger regions of the celestial background. Given their widespread deployment across the Earth's orbital space, they could collectively serve as an opportunistic, distributed SSA network. Operating at frame rates of 2–10 Hz, star trackers can generate large volumes of optical data. However, because they were not originally designed for this purpose, most star-trackers do not store imagery onboard, and many spacecraft lack the downlink capacity required to transmit such large datasets. To our knowledge, no

existing publicly available dataset combines on-orbit, wide-field, star-tracker-class imagery with both RSO annotations and astrometrically calibrated stars, enabling joint RSO detection and spacecraft attitude estimation under real orbital imaging conditions.

While most star trackers do not downlink imagery, an important exception is the Fast Auroral Imager (FAI) aboard the Cascade, Smallsat and Ionospheric Polar Explorer (CASSIOPE) satellite, part of the Enhanced Polar Outflow Probe (e-POP) mission¹⁷. Although designed for auroral science, FAI shares key characteristics with star trackers, including short exposures and wide-field optical sensitivity. Crucially, FAI routinely captures the celestial background with RSOs transiting its field of view¹³, and both imagery and associated ephemeris and attitude data are publicly available through the University of Calgary’s e-POP data portal¹⁸, which is an uncommon provision by spacecraft operators. This accessibility has already led to exploratory demonstrations of FAI as a dual-use sensor for SSA, including applications in RSO detection, orbit and attitude estimation, simulations and wide-field opportunistic mission design studies¹⁹⁻²⁶.

Existing datasets relevant to SSA and spacecraft navigation fall into several categories. Synthetic collections such as SpaceDet²⁷, SPARK²⁸, STAR-24K²⁹ and BUAA-SID-POSE³⁰ provide large volumes of annotated imagery but remain limited by the simplifications inherent to simulated environments. A separate class of resources, including BUAA-SID/POSE, SPEED³¹, SPEED+³², and URSO³³, target satellite pose estimation and six degrees of freedom navigation at close range; while valuable for their intended purposes, these are not designed for wide field of view RSO detection and are therefore not directly comparable to our dataset. Real datasets also exist but are either restricted or lack orbital realism: NEOSSat’s imagery has been leveraged for SSA applications^{34,35} but remains internal to Defence Research & Development Canada (DRDC) and select partners, while RSONAR provides balloon-borne wide FOV star-field imagery³⁶⁻³⁸ that does not capture the complexities of orbital dynamics. Other specialized efforts, such as ST Telemetry³⁹, provide only quaternion attitude data without imagery, and STAR Nav Moon, focus on star tracker calibration or attitude estimation under constrained laboratory or pseudo-operational conditions⁴⁰. Table 1 summarizes key differences among representative datasets.

Dataset	On-Orbit	Wide-FOV	Real Imagery	RSO Labels	Attitude Support	Public
NEOSSat	✓	×	✓	×	×	×
RSONAR	×	✓	✓	✓	✓	×
DUST (ours)	✓	✓	✓	✓	✓	✓

Table 1 | Comparison of DUST with existing space-object imagery datasets.

A checkmark (✓) indicates the dataset provides the corresponding capability; a cross (×) indicates it does not.

To address this gap, we introduce DUST (Dual-Use Star tracker) Dataset⁴¹, a systematically curated collection of 1,378 near-infrared spaceborne images acquired by FAI aboard the CASSIOPE spacecraft between January and August 2023. Each frame has been astrometrically calibrated against catalogued stars, and the dataset includes 4,237 manually verified RSO annotations spanning 160 distinct transits. For each observation, DUST additionally provides spacecraft ephemeris, image-quality metrics, and background noise estimates, enabling analyses that extend beyond object detection to include multi-frame tracking and attitude determination. All imagery and metadata have undergone expert review to ensure annotation accuracy and to establish reliable ground truth. By combining realistic orbital imaging conditions with systematic labelling and supporting telemetry, DUST provides the first openly accessible dataset enabling joint evaluation of RSO detection and orientation inference under operationally realistic conditions.

In this work, we primarily present DUST as a carefully curated and technically validated on orbit dataset, while the included detection and tracking baselines serve only as illustrative examples of its use rather than a comprehensive benchmarking study. DUST provides an openly available dataset that unifies authentic on orbit

imagery with systematic annotations. Its contributions to the space situational awareness and spacecraft navigation communities include:

- **RSO detection in dense star fields:** annotated imagery supports training and evaluation of machine learning-based detection models.
- **Multi-object tracking:** time-sequenced frames enable the linking of detections of the same RSO across consecutive observations.
- **Attitude determination:** provides star tracker imagery, enabling testing of star identification and attitude estimation methods.
- **Challenging real-world imaging conditions:** includes high and spatially varying background noise, stray light and lens flare, spacecraft-induced motion jitter, and extremely small, low-contrast targets characteristic of on-orbit star-tracker imagery.
- **Benchmark resource:** offers one of the first systematically annotated, on-orbit star tracker datasets for SSA, enabling standardized evaluation across detection and tracking methods.
- **Reproducibility:** openly accessible data that supports reproducible algorithm development and cross-domain studies relevant to SSA and astronomy.

Methods

To construct the DUST dataset, a systematic pipeline was developed for data acquisition, preprocessing, annotation, and quality assessment (Fig. 1). The workflow begins with the retrieval of imagery and ancillary metadata from the e-POP FAI, followed by star detection and astrometric calibration, background modeling, and RSO annotation.

Satellite platform and instrument. The CASSIOPE operates in an elliptical low Earth orbit (perigee ≈ 325 km, apogee ≈ 1500 km, inclination 81°) and carries the e-POP payload. The FAI, one of the e-POP instruments, is a dual-channel charge-coupled device (CCD) system designed for wide-field auroral observations. The near-infrared channel (650–1100 nm) was used for this dataset, providing a 26° field of view through a 68.9 mm f/4 lens onto a 256×256 -pixel CCD ($26 \mu\text{m}$ pixels; $\approx 382''/\text{px}$). The detector exhibits a quantum efficiency of ≈ 0.66 in the NIR band. Exposures are 0.1 s at 1 Hz cadence, producing 16-bit images. The mission archive distributes Level 1 products after preprocessing steps, including dark current subtraction, flat-fielding, and distortion correction. Spacecraft attitude and ephemeris metadata are embedded in the image headers.

Image Acquisition and Session Selection. Images used in the DUST dataset were retrieved from the e-POP FAI¹⁸ archive during 2023 in PNG format (256×340 pixels). Each PNG included a header strip outside the active imaging region, which was cropped to recover the native 256×256 -pixel frame. Metadata were extracted from the associated Level-1 Hierarchical Data Format (HDF5) files, which provide frame-synchronized Coordinated Universal Time (UTC) timestamps and spacecraft state vectors, including geographic position, attitude (yaw, pitch, roll), and CCD temperature. Metadata were parsed using the h5py Python library⁴² and aligned frame-by-frame using UTC timestamps. Imaging sessions were included if they provided continuous high-resolution NIR coverage with at least one detectable RSO transit. Based on this criterion, 22 turn-on sessions acquired between January and August 2023 were selected, yielding 1,378 frames in total. Sessions ranged from 24 to 139 frames, depending on visibility and image quality. Their geographical distribution is shown in Figure 3a, and cumulative stellar sky coverage is shown in Figure 3b.

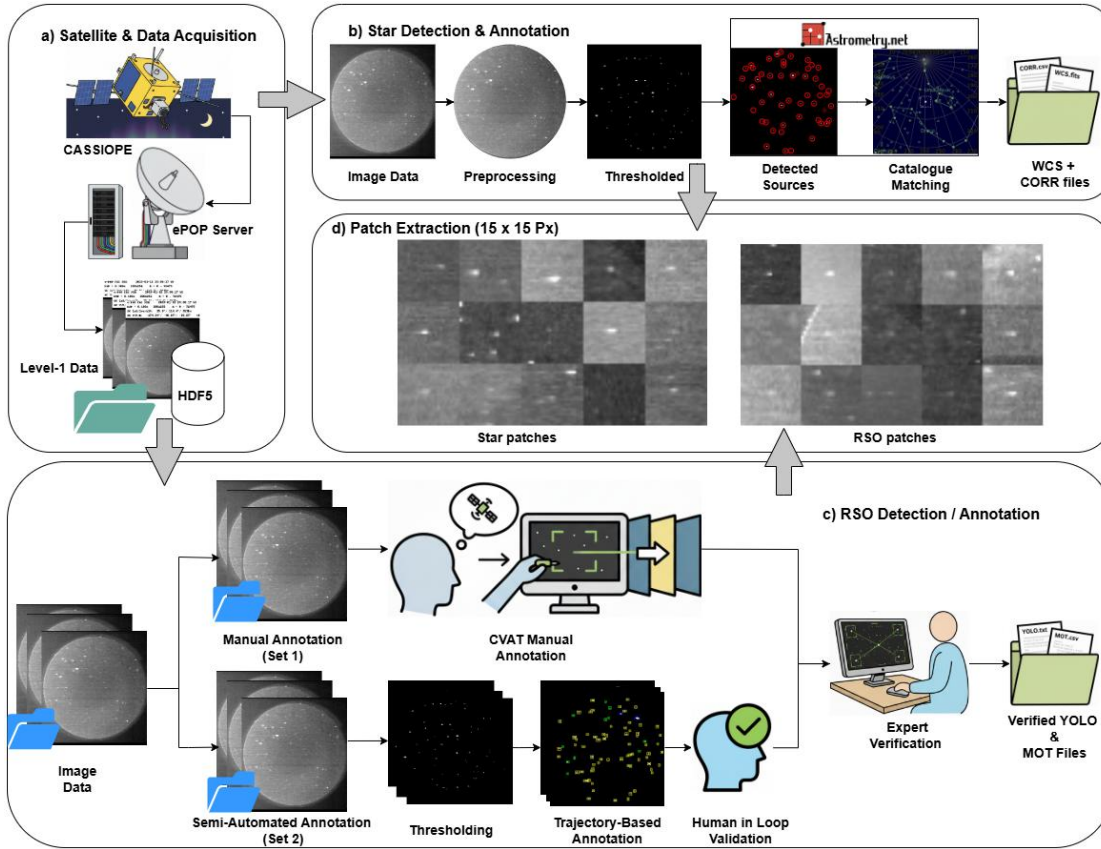


Figure 1 | Data-generation pipeline for the DUST dataset. (a) Acquisition of Level-1 imagery and metadata from the FAI aboard CASSIOPE. (b) Star detection via preprocessing, thresholding, and catalogue matching with Astrometry.net, yielding WCS calibration and correspondence files. (c) RSO detection and annotation using manual and semi-automated workflows with expert validation, exported in YOLO and MOT formats. (d) Patch extraction of 15×15-pixel crops around stars (left) and RSOs (right), supporting classification.

Star Detection and Annotation. Stars were annotated in the dataset by first isolating point-like stellar sources from the image background (Fig. 1b). Pixels outside the instrument’s usable circular field of view were excluded with a 128-pixel radius mask centered on each frame. To further suppress stray light near the field boundary, the images were subsequently cropped to a 120-pixel-radius central region. To account for spatially varying illumination (e.g., vignetting, lens flare, and gradients), a local adaptive thresholding approach was applied. Each frame was divided into 32×32 -pixel tiles, and thresholds were computed independently for each tile. Bright pixels were segmented using a multi-level Otsu algorithm⁴³; if this failed due to a limited dynamic range, a fallback single-level Otsu method with histogram control was applied. The resulting slice-wise masks were aggregated to form full-frame binary masks of stellar candidates.

These masks were submitted to the Astrometry.net⁴⁴ solver, which performs blind astrometric calibration by matching detected point patterns to pre-indexed star catalogues. The solver returned full World Coordinate System (WCS) solutions, including pixel-to-sky transformations, and a correspondence file for each image. The correspondence files list each detected star with its image-plane centroid (x, y), the matched catalogue Right Ascension (RA) and Declination (Dec), Tycho-2⁴⁵ VT magnitude, and a match confidence score. These annotations constitute the stellar component of the dataset, enabling both image-plane analysis and cross-referencing with external catalogues.

During the 20 June 2023 session, strong lens flare initially prevented plate solving. In this case, global thresholding was replaced with a local 21×21 -pixel statistic: pixels were flagged when they exceeded the local

median by three local standard deviations (computed with `scipy.ndimage.generic_filter`)⁴⁶. The resulting candidates were then expanded with two iterations of morphological dilation (`scipy.ndimage.binary_dilation`)⁴⁶ to produce more contiguous masks. These masks were then manually cleaned of spurious streaks using Microsoft Photos “Generative Erase”. Edits were applied only to the dilated binary masks, not the original images. Both pre- and post-correction masks are included in the dataset under the `/Modified_Images/` directory.

RSO Annotation. RSOs were annotated using a combination of manual and semi-automated workflows to balance accuracy and efficiency (Fig. 1c). Sessions were divided into Set 1 (manual) and Set 2 (semi-automated), as summarized in Table 2 of the Data Records. The use of two annotation methods reflects the sequential development of the labelling pipeline. Set 1 was annotated prior to the development of the semi-automated suite using CVAT⁴⁷, providing a human-labelled reference; annotations were generated frame-by-frame with outputs saved in YOLO format (bounding box centre x, y ; width; height; class label, normalized to image dimensions).

Set 2 was annotated using a custom four-module Python annotation suite²⁶, enabling more efficient processing of the larger session volume while maintaining expert oversight. Binary detection masks were first generated per frame using a windowed multi-Otsu thresholding algorithm (`mask_maker`), then refined through a brush-based pixel editing interface overlaid on the original image (`mask_editor`). Candidate objects were subsequently identified from the refined masks using a Connected Component Algorithm (CCA), which extracted centroid coordinates, bounding box dimensions, and pixel intensity for each detection (`object_classifier`). Candidates were linked across frames using a nearest-neighbour motion model, with inter-frame displacement used to distinguish RSOs from stars; RSOs traverse the field of view with measurably different angular velocities relative to the quasi-static stellar background. Bounding boxes and class labels were output in YOLO format. A final interactive GUI (`object_class_editor`) enabled expert review, where an annotator verified track persistence, removed false positives, and corrected misclassifications on a per-object basis.

Regardless of annotation method, all sessions in both sets were reviewed by an SSA domain expert to verify annotation quality, flag errors, and ensure consistency across the dataset. The two methods nonetheless produce systematically different bounding box extents by design: manual annotation tends to encompass objects with a small margin, while CCA-derived boxes are bounded tightly by the detected pixel mask. This inter-set difference is acknowledged, quantified, and discussed in the Technical Validation section, where per-set bounding box distributions, morphological robustness under dilation and erosion, and practical guidance for users combining both sets are provided. Final annotations are distributed in two formats: (i) YOLO, for object detection, and (ii) MOT (Multiple Object Tracking), which records frame ID, track ID, and bounding box parameters in pixels. Annotation continuity and consistency across frames were further assessed in the Technical Validation section.

Background Characterization. Background properties were quantified to assess image quality and identify systematic illumination artifacts. For each frame, binary source masks were inverted to isolate background pixels, from which the median intensity, the arithmetic mean, standard deviation, and a robust dispersion estimate based on the median absolute deviation (MAD, scaled by 1.4826) were computed. These complementary metrics describe both central tendency and variability, with the MAD-based σ offering robustness to outliers such as hot pixels or residual streaks.

To characterize large-scale gradients, images were transformed into polar coordinates (radius–azimuth) with the optical center as origin. Pixel intensities were binned into radial–azimuthal sectors and averaged across all frames in a session to generate two-dimensional polar illumination maps. From these, one-dimensional radial profiles were derived by azimuthal averaging and normalized to the 99th percentile brightness for cross-session comparison. A 128-pixel circular mask restricted analysis to the usable field of view. Together, these measures capture both fine-scale noise properties and structured illumination artifacts such as lens flare, stray light, or diffuse Galactic background.

Baseline Feature Separability. In the FAI imagery, both stars and RSOs appear as near point-like sources, making them visually difficult to distinguish. To test whether image features provide separability, 15×15 -pixel crops were extracted around star centroids (from correspondence files) and RSO centroids (from MOT files), as shown in Fig. 1d. Crops extending beyond image boundaries were excluded.

First, ten handcrafted features were computed directly at native resolution (15×15 px, no interpolation) from each grayscale patch. Let I_{ij} denote the pixel intensity at row i , column j , with $dx_j = x_j - W/2$ and $dy_i = y_i - H/2$ denoting pixel offsets from the patch centre, where W and H are the patch width and height in pixels. A small regularisation term $\varepsilon = 10^{-6}$ is added to all denominators and under the square root for numerical stability. The total regularised intensity is:

$$w_{\text{tot}} = \sum_{i,j} (I_{ij} + \varepsilon) \quad (1)$$

The intensity-weighted second-order moments are:

$$M_{xx} = \frac{\sum_{i,j} (I_{ij} + \varepsilon) dx_j^2}{w_{\text{tot}}}, \quad M_{yy} = \frac{\sum_{i,j} (I_{ij} + \varepsilon) dy_i^2}{w_{\text{tot}}} \quad (2)$$

Elongation is defined as the ratio of eigenvalues $\lambda_{\text{major}}/\lambda_{\text{minor}}$ of the symmetric moment tensor, whose off-diagonal term $M_{xy} = \sum (I_{ij} + \varepsilon) dx_j dy_i / w_{\text{tot}}$ is used solely as an intermediate quantity, with λ_{minor} clamped to a minimum of 10^{-9} to prevent division by zero in degenerate patches:

$$\lambda = \frac{M_{xx} + M_{yy}}{2} \pm \sqrt{\left(\frac{M_{xx} - M_{yy}}{2}\right)^2 + M_{xy}^2} \quad (3)$$

The FWHM is estimated as:

$$\text{FWHM} = 2.355 \sqrt{\frac{M_{xx} + M_{yy}}{2} + \varepsilon} \quad (4)$$

The concentration index is defined as the ratio of flux within a radial distance $r \leq 3$ px from the patch centre to the total flux:

$$\text{Concentration} = \frac{\sum_{r_{ij} \leq 3} I_{ij}}{\sum_{i,j} I_{ij} + \varepsilon}, \quad r_{ij} = \sqrt{dx_j^2 + dy_i^2} \quad (5)$$

The remaining five features are: peak intensity ($\max(I_{ij})/255$), mean intensity ($\text{mean}(I_{ij})/255$), signal-to-noise ratio $\text{SNR} = (\max(I_{ij}) - \text{median}(I_{ij})) / (\sigma + \varepsilon)$ where σ is the pixel standard deviation, background-subtracted flux = $(\text{mean}(I_{ij}) - \text{median}(I_{ij})) / 255$, and excess radial kurtosis, defined as the fourth standardised moment minus 3 computed over pixel intensities sorted by ascending radial distance r_{ij} from the patch centre. All ten features were computed on the grayscale patch without any upscaling or interpolation.

Second, each crop were replicated across three channels, resized to 224×224 pixels using the ResNet-50⁴⁸ preprocessing pipeline, and passed through a ResNet-50 model pretrained on ImageNet-1K_V2⁴⁸. The final fully connected layer was replaced with an identity mapping to yield 2048-dimensional embeddings, with no fine-tuning. Crops extending beyond image boundaries were excluded.

Each class was capped at 3,000 samples using a fixed random seed of 42 to maintain balance; actual yields after boundary exclusion were 2,976-star patches and 2,960 RSO patches. Up to 50,000 random pairs per category (star-star, RSO-RSO, star-RSO) were sampled for each feature representation to compute pairwise cosine distance distributions ($d = 1 - \cos \theta$); self-pairs were excluded from within-class sampling. Pairwise cosine distances were computed on L2-normalised raw features, as cosine similarity is invariant to per-vector scaling.

Both feature sets were standardised (z-scored) prior to dimensionality reduction and classification. Native features were projected into two dimensions using UMAP⁴⁹ with cosine distance (15 neighbours, minimum distance 0.10, random seed 42). Linear separability was assessed for each representation by training a logistic regression classifier (L2 penalty, Limited-memory Broyden–Fletcher–Goldfarb–Shanno⁵⁰ solver (lbfgs), maximum 500 iterations for native features and 2,000 for ResNet-50 embeddings, class-balanced weighting) on a stratified 70/30 train–test split (random seed 42).

Data Records

The DUST dataset is publicly available via Zenodo at <https://doi.org/10.5281/zenodo.20255672>⁴¹, under CC-BY 4.0 as of May 17, 2026.

The dataset is provided in open formats (PNG, CSV, TXT, and FITS). Files are arranged hierarchically by annotation set (set 1 = manual, set 2 = semi-automated) and by observation date (YYYY_MM_DD), with duplicate dates suffixed a or b where multiple sessions occurred on the same night. All files are indexed by ISO 8601 timestamps for consistent cross-referencing. Set 1 and Set 2 share the same folder hierarchy and naming conventions. The directory structure is illustrated in Figure 2. A per-session index is provided in Table 2.

Each session directory contains a consistent set of products generated during data processing:

- **Background/** —One CSV per session. Columns: Filename; Median (DN); MAD-based Sigma (DN); Mean (DN); Std Dev (DN).
- **Corr/** —One CSV per successfully plate-solved frame, co-named with the corresponding image. Not all frames yield a WCS solution; sessions with low star counts or severe background contamination may have no corresponding Corr file. Columns: field_x, field_y (detected star centroid, pixels); field_ra, field_dec (RA/Dec, degrees J2000); index_x, index_y (catalogue position, pixels); index_ra, index_dec (catalogue RA/Dec, degrees J2000); index_id (catalogue identifier); field_id (field source identifier); match_weight (match confidence, 0–1); MAG_BT, MAG_VT, MAG_HP (Tycho-2 BT, VT and Hipparcos Hp magnitudes); MAG (best magnitude); FLUX (source flux, DN); BACKGROUND (local background, DN).
- **H5/** —One HDF5 file per frame, co-named with the corresponding image, containing one dataset (FAI Image, 280 × 256, float64) and 20 root-level metadata attributes: UTC; S/C Latitude (°), S/C Longitude (°), S/C Altitude (km); Yaw (°), Pitch (°), Roll (°); CCD Temperature (°C); Exposure Time (s); Camera Mode; Camera Source; DN to Rayleigh Conversion Factor (R/DN/s); Magnetic Latitude (°), Magnetic Longitude (°), Magnetic Local Time; S/C GEO X/Y/Z Position (km); S/C GSM X/Y/Z Position (km); Attitude Determination Mode; Image Projection Available; Source File Name; Version.

Table 2 | Observation sessions included in the DUST dataset. Each row lists the session date, UTC observation window, number of frames, total and unique RSOs, stars detected, and annotation method (Set 1 = manual, Set 2 = semi-automated)

Date	Observation window (UTC)	Frames	RSOs (total)	RSOs (unique)	Stars (total)	Stars (unique)	Annotation set
2023-01-16	21:27:05 – 21:27:45	41	41	1	1651	120	1
2023-01-21	22:48:31 – 22:49:46	76	60	3	2679	161	1
2023-01-25	21:44:17 – 21:45:11	53	88	4	1147	89	1
2023-02-21	08:18:14 – 08:19:03	50	263	15	1044	74	2
2023-03-16	06:30:14 – 06:31:03	50	119	4	1074	81	2
2023-03-26	07:40:14 – 07:41:04	50	160	5	366	61	2
2023-03-31	16:20:14 – 16:22:33	139	450	13	2722	184	1
2023-04-22	05:37:14 – 05:38:03	50	364	13	1140	91	2
2023-05-08	21:03:14 – 21:04:03	50	86	4	1600	120	2
2023-05-10	07:46:15 – 07:47:05	50	57	2	1605	120	2
2023-05-31	10:56:14 – 10:58:22	128	422	17	3421	172	1
2023-06-03	00:11:58 – 00:13:44	91	430	12	1908	117	1
2023-06-05	03:27:14 – 03:28:03	50	142	3	1605	120	2
2023-06-19	00:51:14 – 00:52:03	50	427	15	2071	144	2
2023-06-20	00:52:05 – 00:52:43	39	61	2	161	70	1
2023-07-19	03:33:54 – 03:35:47	114	276	8	3308	200	1
2023-07-20	01:02:14 – 01:03:03	50	62	5	1885	116	2
2023-07-21	00:05:14 – 00:06:03	50	366	16	1547	121	2
2023-08-04	21:05:42 – 21:06:05	24	38	2	809	97	1
2023-08-04	21:08:16 – 21:09:30	75	185	8	2115	134	1
2023-08-05	02:06:41 – 02:07:59	35	70	3	1196	156	1
2023-08-05	23:36:14 – 23:37:16	63	70	5	823	76	1

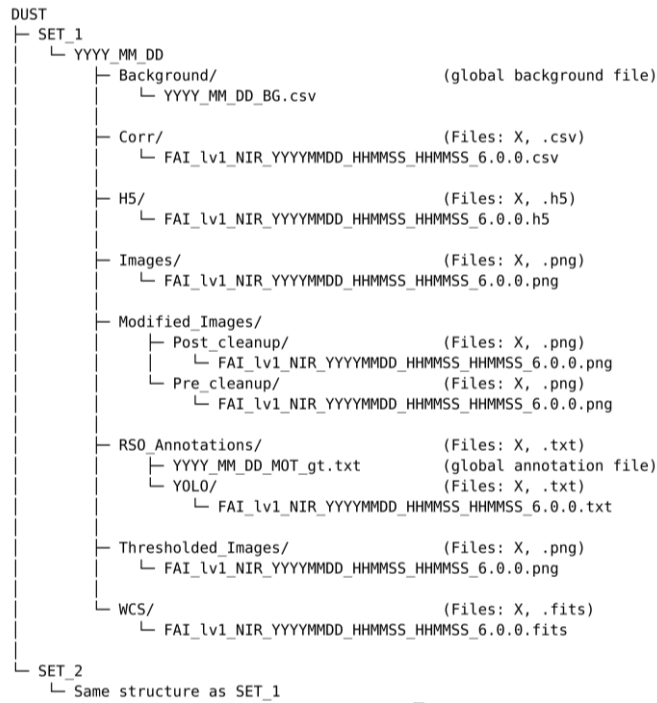


Figure 2 | Directory hierarchy and file organization in DUST. The figure summarizes the standardized structure used across all observation sessions, illustrating how data products are grouped by annotation source and acquisition date.

- **Images/** — One 256×256 px, 8-bit PNG per frame, named by UTC timestamp (YYYY_MM_DD_HH_MM_SS.png), containing calibrated FAI near-infrared (650–1100 nm) exposures (pixel values 0–255 DN).
- **Modified_Images/** — Present in the 2023-06-20 session only. Pre- and post-cleanup binary masks distinguished by `_pre` and `_post` filename suffixes.
- **RSO_Annotations/** — Labels in two formats. Per-frame YOLO TXT files (space-delimited, no header) are stored in a YOLO/ subfolder; each row contains: class (RSO = 1); `x_centre`, `y_centre`, width, height (all normalised 0–1). A session-level MOT file (YYYY_MM_DD_MOT_gt.txt, comma-delimited, no header) contains frame (1-indexed); track_id; x, y, w, h (floating-point pixels, top-left origin); conf (always 1); class (RSO = 1); visibility (always 1.0). Note: RSOs are class 1 in both formats.
- **Thresholded_Images/** — One 256×256 px, uint8 PNG per frame, co-named with the corresponding image. Near-white pixels (≈ 255) indicate detected sources; near-zero pixels (≈ 0) indicate background. .
- **WCS/** — One FITS file per successfully plate-solved frame containing the Astrometry.net World Coordinate System solution encoding the pixel-to-sky transformation.

Data Overview

DUST comprises 1,378 near-infrared image frames (256×256 px, 1 Hz cadence) acquired across 22 observation sessions between January and August 2023. The dataset contains 4,237 annotated RSO instances spanning 160 distinct transits. Twelve sessions were annotated manually (Set 1, 878 frames) and ten semi-automatically (Set 2, 500 frames). The dataset additionally provides per-frame astrometric solutions, spacecraft ephemeris and attitude, and background noise profiles. A per-session breakdown is provided in Table 2.

Technical Validation

Space Attitude Stability. Attitude stability was assessed by analyzing time-resolved yaw, pitch, and roll angular rates. Prior to analysis, angles were unwrapped to remove discontinuities at $\pm 180^\circ$, ensuring that rate estimates

were not biased by artificial jumps. Most sessions exhibited uniform yaw-dominated slews with rates of 2520–3960 arcsec s⁻¹ and dispersions below 250 arcsec s⁻¹, confirming stable dynamics with low jitter. Step-like transitions occasionally observed in the profiles reflect expected control events, such as the termination of slews, and do not indicate instability. One session (2023-05-31) diverged from this pattern, showing elevated dispersions and excursions above 36,000 arcsec s⁻¹ consistent with a maneuver; this was retained for completeness but excluded from aggregate statistics. Overall, these results confirm reproducible pointing stability across sessions, supporting the dataset’s use in dynamic imaging and attitude studies.

Star Detection and Astrometric Accuracy. The star detection and plate-solving pipeline was evaluated for detection yield, coverage, match reliability, astrometric accuracy, and brightness. Adaptive thresholding combined with Astrometry.net achieved an overall plate-solving success rate of 84% (median 92%). Most sequences consistently solved at 100%, while failures below 60% were confined to background or presence of an artefact. In total, 35,877 stars were detected with 2,624 unique stars (see table 2), with 8–69 per frame (median 30), depending on conditions. Stars were detected across 38 constellations, though nearly half of detections lie within Sagittarius, Virgo, Scorpius, Capricornus, and Gemini. Coverage is concentrated at Galactic latitudes $|b| \lesssim 45^\circ$ (angular distance from the Galactic plane), with enhanced density around $|b| \approx 15\text{--}30^\circ$. Very few pointings occur at $|b| > 60^\circ$, highlighting clustering toward mid-Galactic latitudes. The cumulative sky coverage is shown in Figure 3b. The dataset samples $\approx 0.04\%$ of the celestial sphere at 382"/px resolution, with 1,513 distinct regions identified and 74% revisited (median depth seven frames).

Match quality assessed using the Astrometry.net match-weight statistic (0–1), had a median value of 0.95 and a 90th-percentile value of 0.988. Astrometric accuracy was sub-pixel relative to the 382"/px scale. After IQR filtering, 68%, 90%, and 95% of detections aligned within 240.5" (0.630 px), 385.7" (1.010 px), and 485.4" (1.271 px), respectively (see figure 5c). Residuals were isotropic, centered near zero, and stable across sessions (medians 162.6–275.0"). A mild increase with field radius was observed (~ 0.2 px from center to edge), consistent with edge-of-field distortions. Brightness distributions derived from Tycho-2 magnitudes, with peaks at 4–6 mag and decline beyond ~ 6.5 mag, reflecting thresholding effects. Although segmentation was not strictly photometric, the distributions are robust and reproducible, supporting downstream applications. Together, these results demonstrate that the dataset provides reliable, well-calibrated stellar detections suitable for cross-validation, benchmarking of astrometric and object-detection pipelines, and as training or evaluation data for machine-learning models operating on space-based optical imagery.

RSO Detection and Annotation. Across the dataset, RSOs occupy compact bounding boxes, with widths of 2.0–6.5 px (5th–95th percentile; median 4.0 px) and heights of 1.0–5.25 px (median 3.1 px). Areas span $\sim 3.0\text{--}31.84$ px², with a median of 12.6 px². Occasional long tails reflect bright RSOs, but most detections are tightly clustered, consistent with near point-like sources. At the dataset’s angular scale (≈ 382 arcsec/px), these correspond to apparent sizes of $\sim 1\text{--}3$ arcmin, matching expectations for short-exposure (0.1 s) imagery. Manual (Set 1) and semi-automated (Set 2) annotations show systematic differences: Set 1 boxes are larger (median ≈ 18 px²) while Set 2 boxes are tighter (median ≈ 7 px²), with the largest divergence arising from box height (median 3.94 px vs. 2.0 px). This reflects the effect of thresholding and the greater variability of manual annotation at the pixel scale. Violin plots (figure 5d) confirm this shift, with Set 2 distributions concentrated in the 1–3 px regime

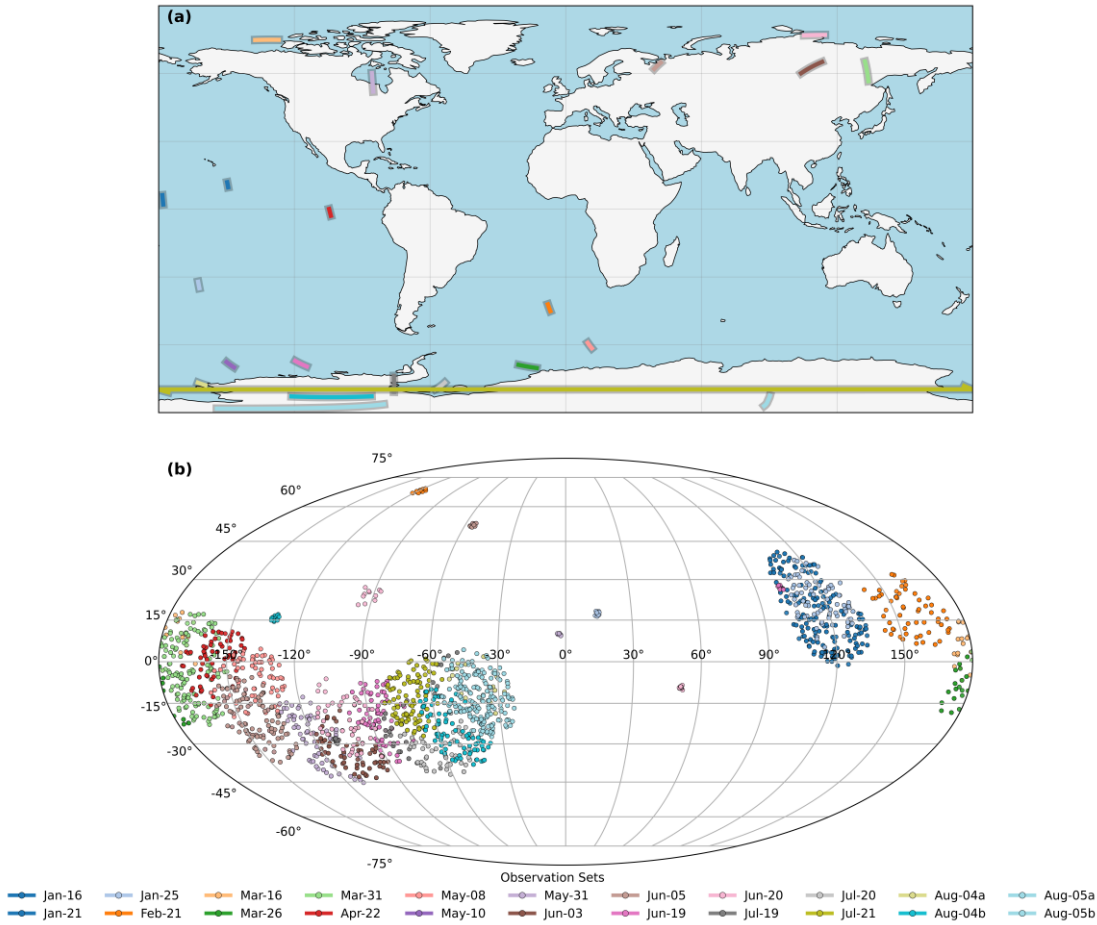


Figure 3 | Geographical and celestial coverage of the DUST dataset. (a) Ground-track footprints of the 22 observation sessions (January–August 2023) in geographic coordinates. (b) Mollweide projection of the celestial sphere showing the distribution of plate-solved stars. Colors indicate individual sessions, consistent across both panels. Together, the maps illustrate the combined Earth-track and sky coverage of the dataset.

The dataset is dominated by “tiny” objects in computer vision terms: $\sim 76.8\%$ of RSOs are width ≤ 5 px and height ≤ 5 px, and $\sim 85\%$ are ≤ 25 px². By comparison, MS COCO⁵¹ defines “small” as $\leq 32 \times 32$ px, while AI-TOD⁵² subdivides very tiny (2–8 px) and tiny (8–16 px). Thus, this dataset sits at the extreme end of difficulty, with many RSOs near the very tiny threshold. Morphological ± 1 px tests revealed strong sensitivity to pixel-level perturbations: Set 1 median areas shifted by $+116.5\%/-72.1\%$ and Set 2 by $+242.9\%/-71.4\%$. While 68.1% of Set 2 RSOs remained ≤ 25 px² under all conditions, and only 32.9% of the full dataset stayed within the tiny-object regime after dilation and erosion. These results confirm that many RSOs are intrinsically tiny, though annotation boundaries are highly sensitive at 1-px scales.

Track Completeness and Coherence. Per-object track completeness, defined as the fraction of frames in which each RSO was annotated, averaged 98.4% (IQR 97.1–99.5%). Only two sessions ($<2\%$) fell below 90%, primarily due to transient occlusions or low-SNR conditions. Tracks were temporally coherent, with smooth centroid motion, stable bounding boxes, and no evidence of ID switching; ID consistency was verified manually by plotting each RSO's trajectory across every frame in a sequence and visually inspecting for label swaps during annotation. Occasional gaps corresponded to expected flux dropouts (figure 4). Following the 2021 loss of reaction wheels, the spacecraft adopted a spin-stabilized, anti-sun pointing mode⁵³. This configuration produces curved trajectories for both stars and RSOs, rather than strictly linear motion. This realistic orbital context further enhances the dataset's value for testing robust tracking algorithms.

Per-track stability was quantified using velocity standard deviation (σ_v) and bounding box IoU variability (σ_{IoU}). Median dataset-wide values were $\sigma_v = 0.58$ and $\sigma_{IoU} = 0.0$. Sessions spanned four stability regimes: (i) smooth and stable (18.2%), (ii) smooth but shape-variable (22.7%), (iii) stable boxes with jittery motion (36.4%), and (iv) both unstable (22.7%). The most challenging cases involved faint RSOs near detection thresholds. Even in these conditions, tracks remained coherent and no ID switching was observed under manual visual inspection. For machine learning, these quadrant distinctions highlight which sessions may introduce label noise or serve as stress tests for robust tracking. These metrics demonstrate that while most sessions are high-quality, the dataset also includes controlled variability, making it valuable for benchmarking tracking methods under realistic noise and detection limits.

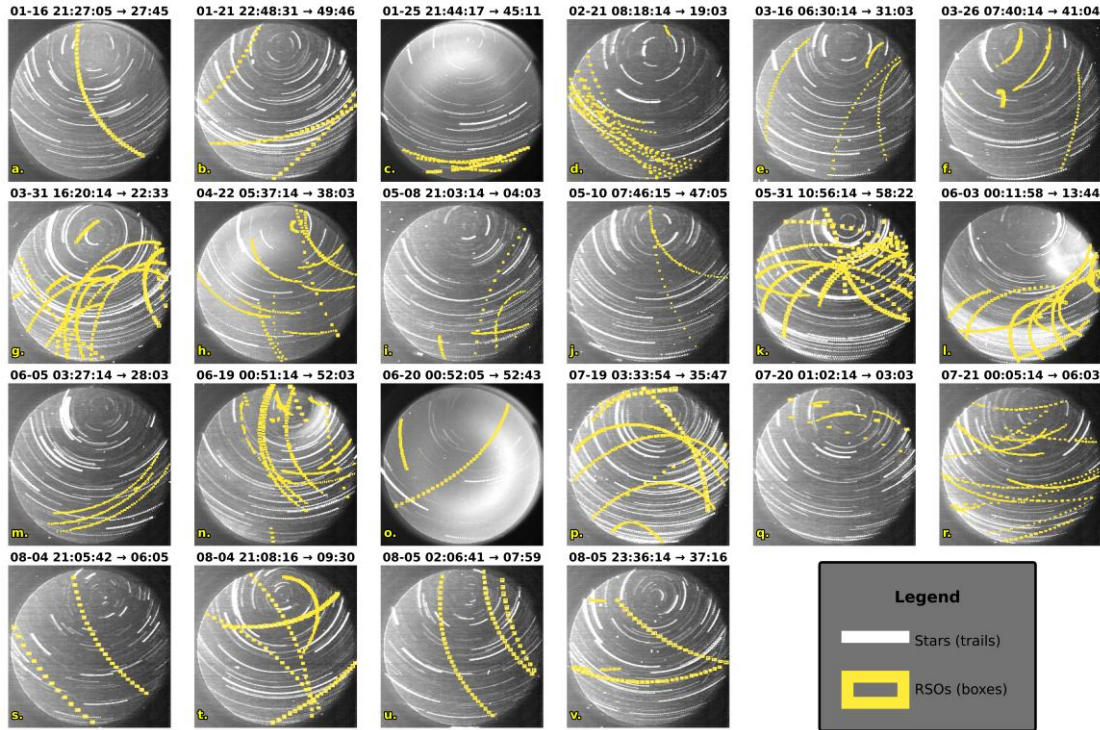


Figure 4 | Annotated RSOs across all observation sessions. Stacked images from each of the 22 observation sessions (January–August 2023), showing the FAI’s circular field of view. Stellar trails are visible in white, while RSOs are highlighted with yellow bounding boxes. Session timestamps (UTC) are given above each panel. The montage demonstrates the consistency of RSO annotations across diverse observing conditions, including dense stellar fields, faint backgrounds, flares, and stray light.

Spatial Priors in Object Distributions. Two-dimensional kernel density estimates (KDEs) were computed for image-plane centroids: stars from plate-solved detections and RSOs from bounding-box centers. Star centroids were nearly uniform across the field of view (figure 5a), with small but significant deviations ($\chi^2 = 27.7$, $p = 4.14 \times 10^{-6}$) attributable to mild vignetting. In contrast, RSO centroids exhibited a modest anisotropy (Top = 25.1%, Right = 26.5%, Bottom = 25.3%, Left = 23.1%; $\chi^2 = 9.83$, $p = 0.020$), with a +1.5% point enrichment in the right-hand quadrant, attributable to preferred orbital transit geometries (see figure 5b). These spatial priors are relevant for downstream applications: object detectors may inherit such positional bias, underscoring the value of spatially balanced sampling or region-weighted loss functions during model training.

Baseline Separability of Stars vs. RSOs. For native-resolution (15×15 -pixel) features, pairwise cosine distances were nearly indistinguishable across categories: median distances of 0.048 (Q25–Q75: 0.010–0.139) for star–star pairs, 0.047 (0.011–0.135) for RSO–RSO pairs, and 0.052 (0.012–0.147) for star–RSO pairs, a pattern

consistent with the UMAP projection of the native feature space (Fig. 5e), where stars and RSOs occupy largely overlapping regions with no discernible cluster boundaries. For ResNet-50 embeddings (224×224 pixels patches), the same pattern of within-class and between-class indistinguishability held: median cosine distances of 0.150 (0.109–0.207), 0.146 (0.107–0.200), and 0.152 (0.111–0.210) for star–star, RSO–RSO, and star–RSO pairs respectively. The higher absolute distances in the ResNet-50 space relative to native features are expected given the substantially higher dimensionality (2,048 vs 10), and do not reflect greater separability, within-class and between-class distributions remain equally indistinguishable in both spaces, confirming that off-the-shelf ImageNet features do not naturally separate stars from RSOs in star tracker imagery.

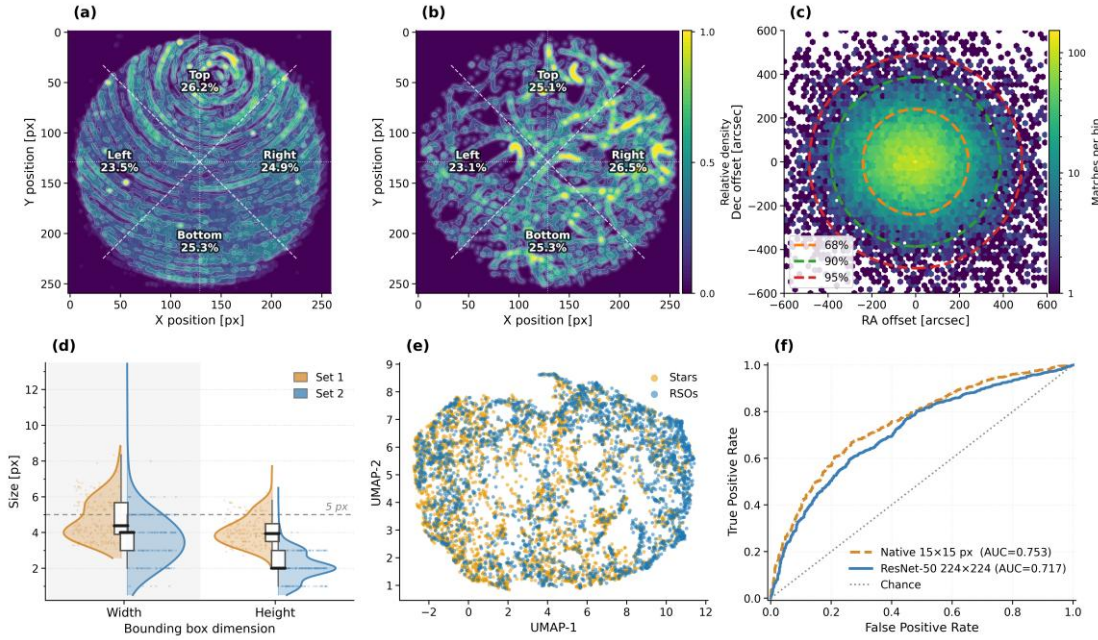


Figure 5 | Technical validation of FAI dataset. (a–b) Spatial density of star centroids and RSOs with quadrant fractions. (c) Astrometric residuals with 68/90/95% containment. (d) Bounding box size distributions across sets. (e) UMAP projection of native 15×15 -pixel handcrafted features (stars orange $n = 2,976$; RSOs blue $n = 2,960$) showing strong class overlap (pairwise cosine distances: star–star 0.048, RSO–RSO 0.047, star–RSO 0.052). (f) ROC curves for native features (orange dashed, $AUC = 0.753$) and ResNet-50 embeddings (blue solid, $AUC = 0.717$); both confirm weak static separability above chance (amber dotted), with the lower native AUC ruling out upscaling as a confound.

A logistic regression linear on native features achieved ROC–AUC = 0.753 and accuracy = 0.696 (confusion matrix: TN = 668, FP = 225, FN = 316, TP = 572). ResNet-50 embeddings achieved ROC–AUC = 0.717 and accuracy = 0.656 (TN = 604, FP = 296, FN = 323, TP = 577). Errors were distributed across both classes in both cases, confirming results are not driven by class imbalance. However, native features achieved modestly higher separability than ResNet-50 embeddings when both were evaluated on standardised features, confirming that the $15\times$ upscaling introduces no spurious discriminative signal and that deep ImageNet embeddings offer no advantage over simple morphological descriptors for distinguishing between the stars and RSOs. Both representations yield only weak linear discriminability ($AUC = 0.717–0.753$) (shown in Figure. 5f), confirming that limited separability reflects genuine intrinsic similarity in static appearance rather than any methodological artefact. Effective discrimination in star tracker imagery therefore requires domain-specific cues including temporal motion and multi-frame context, which DUST is specifically designed to provide.

Image Quality Assessment and Background Modelling. Median background values clustered around a global reference of 101 DN, with 77% of sessions falling within $\pm 10\%$ (Fig. 6a). Two sessions diverged sharply from this pattern: 2023-05-31 was substantially dimmer (-27%), while 2023-06-20 was markedly brighter ($+31\%$) due to lens flare. Apart from these outliers, illumination was stable across most nights.

Temporal stability was evaluated by fitting linear slopes to per-frame medians (Fig. 6b). 8 of 22 sessions (36%) exhibited monotonic drift above the threshold of $0.15 \text{ DN frame}^{-1}$, with slopes ranging from $+0.17$ to $+0.48 \text{ DN frame}^{-1}$, except for one negative drift ($-0.19 \text{ DN frame}^{-1}$). The remaining 64% of sessions showed negligible drift, indicating generally stable background levels over time. Background variability, measured using both standard deviation and MAD-based σ (Fig. 6c), showed strong correlation at both per-frame ($\rho = 0.80$) and per-session ($\rho = 0.80$) levels. Global variability clustered near 40 DN, though the quietest and noisiest sessions differed by metric: 2023-05-31 had the lowest standard deviation (-18%), while 2023-06-03 had the lowest MAD- σ (-22%). Conversely, 2023-06-20 was the noisiest session by standard deviation ($+58\%$) and 2023-01-25 was the noisiest by MAD- σ ($+37\%$). Because MAD- σ is more robust to optical artefacts such as flares, it provides a more stable descriptor of intrinsic background variability.

Cross-metric correlations revealed that median background level was moderately associated with variability ($\rho = 0.53\text{--}0.68$), while drift was largely independent of all other metrics ($|\rho| = 0.43\text{--}0.56$). Together, these complementary descriptors provide a multi-dimensional view of background stability and allow researchers to select sessions tailored to their analysis needs. Finally, systematic artefacts were readily distinguished in polar illumination maps and radial profiles (Fig. 6d & e). Flares produced edge-brightened, irregular signatures, stray light generated smooth monotonic brightening, and Milky Way contamination appeared as diffuse, centrally enhanced backgrounds. These artefacts illustrate the variety of illumination structures captured in the dataset.

Usage Notes

The DUST dataset is designed as a shared resource for both space situational awareness (SSA) researchers and machine learning developers. It consists of on-orbit star tracker images annotated for stars and resident space objects (RSOs), enabling a variety of detection and tracking tasks. Key usage scenarios include:

- **Star-RSO discrimination:** The annotated labels allow training detectors or classifiers to distinguish resident space objects from background stars.
- **Multi-scale object detection:** RSOs appear at widely varying sizes and brightness, providing a challenging testbed for scale-invariant detectors (for example, EfficientDet) that must detect both small and large targets.
- **Tracking under motion jitter and occlusion:** Consecutive frames include spacecraft-induced jitter (causing apparent object motion) and occasional overlapping tracks, enabling evaluation of robust multi-object tracking methods (such as OrbitTrack²⁴) under realistic conditions.
- **Illumination normalization and augmentation:** The images exhibit a range of exposure levels and background brightness. The data can be used to develop or test preprocessing and augmentation techniques that normalize lighting variations across scenes.
- **Benchmark Evaluation.** To illustrate use of the dataset, we include baseline performance using two example models. Applying the OrbitTrack pipeline²⁴ to DUST yields high detection precision ($\approx 97.2\%$) and moderate recall ($\approx 74.2\%$), demonstrating that state-of-the-art SSA tracking methods work on these images. We also trained a Roboflow-hosted EfficientDet⁵⁴-style object detector on the dataset; this baseline achieved a mean average precision of 0.61 at an IoU threshold of 0.50 (mAP@50) on the held-out test split, with precision of 0.73 and recall of 0.51, reflecting the difficulty of detecting tiny RSOs in dense star fields. These results show that modern object detection and tracking algorithms can be directly applied to DUST.

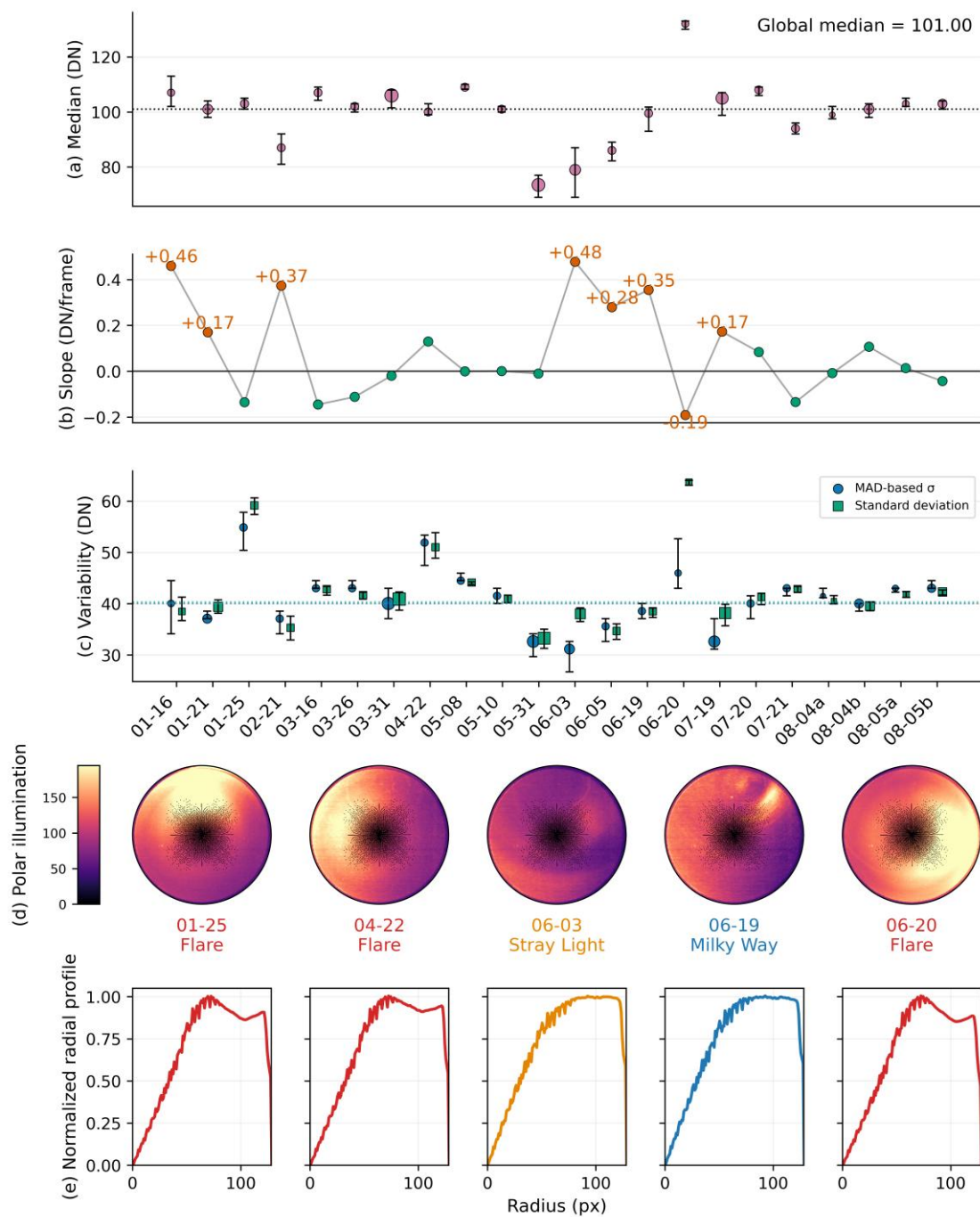


Figure 6 | Session-level background stability and representative illumination artifacts.

(a) Median background values across observation sessions, with interquartile ranges and global reference shown. (b) Drift slopes of background levels within sessions, highlighting temporal trends and outliers. (c) Session-level variability quantified by MAD-based σ and standard deviation. (d) Polar illumination maps illustrating representative sessions with flares, stray light, and Milky Way contamination, with a shared color scale. (e) Normalized radial profiles corresponding to the sessions in (d), showing distinct radial intensity signatures for each artifact type.

Data Availability

The DUST dataset is openly available on Zenodo at <https://doi.org/10.5281/zenodo.20255672> under CC-BY 4.0.

Code Availability

The OrbitTrack algorithm used for RSO detection and tracking with this dataset is openly available at (https://github.com/ausmlab/rso_tracking). The EfficientDet-based baseline configuration used for object-detection benchmarks is available via Roboflow at (<https://universe.roboflow.com/rsoefficientdet/resident-space-objects>). All custom code used for dataset generation, preprocessing, annotation conversion, background characterization, analysis, and figure production is provided alongside the dataset in the associated Zenodo repository at (<https://doi.org/10.5281/zenodo.20255672>).

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Author contributions

V.S.: Writing - original draft, Conceptualization, Writing - Review & Editing, Methodology, Data curation, Software, Investigation, Visualization, Validation, Formal Analysis; P.K.: Writing - Review & Editing, Data curation; R.L.: Funding acquisition, Project Administration, Supervision, Resources, Writing - Review & Editing; G.S.: Writing - Review & Editing, Validation, Visualization.

Competing interests

The authors declare that they have no competing interests.

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